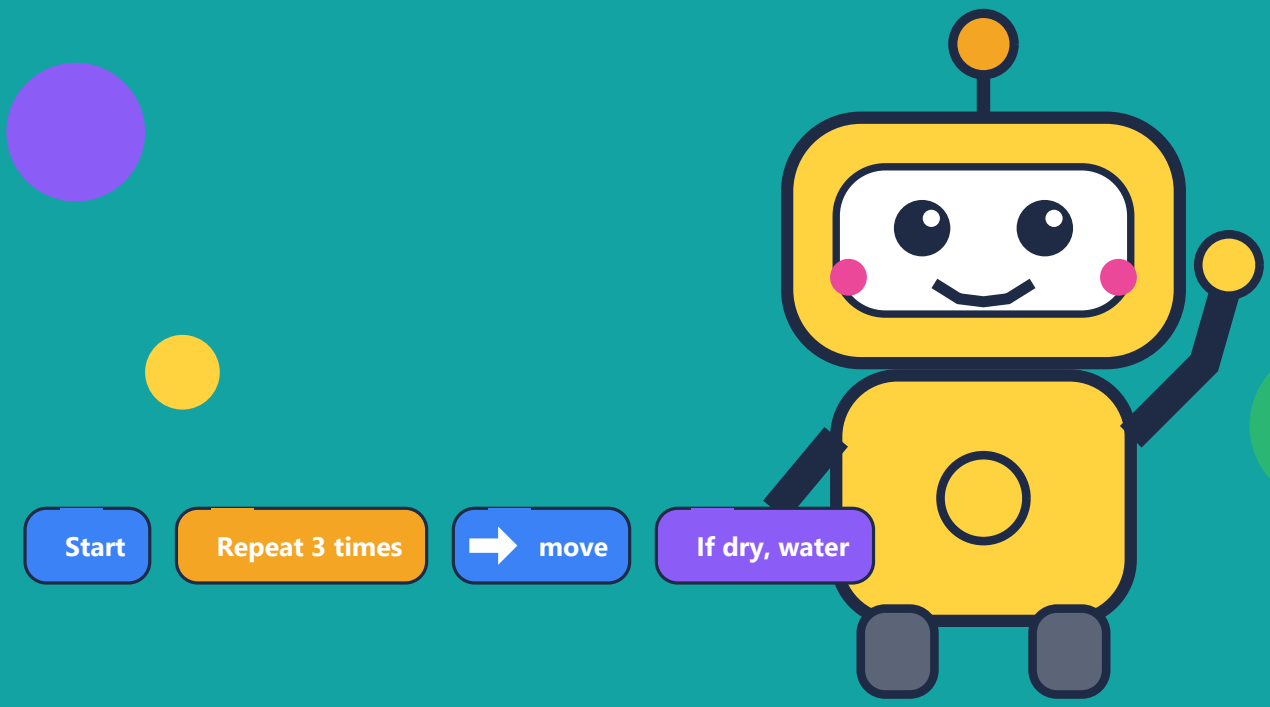
CodeSchool · Class 5

Think Like a Coder

Computational thinking for Class 5

Learn it · See it · Try it · Build it



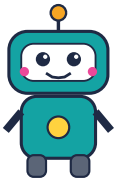
Follows the 15 chapters of the NCERT Class 5 maths textbook

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper and challenges

SAMPLE EDITION · DRAFT

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Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn	Robo explains one big idea in simple words, with the new coder words to remember.
Example	A worked example shows the idea step by step.
Try it!	Five practice questions. Circle the answer or write it in the box.
Now you build it	Coding on paper: put code cards in order, fix bugs and write your own programs.
Challenge	A harder puzzle for when you are ready for more.
How did I do?	Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this **Repeat: a loop** **If: a decision** **When: an event** **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 5 maths textbook, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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Coder skills in this book

Levels: B4 reviewed; B5 variables and inputs; B6 search and decomposition strategies; B7 capstone game design.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	True, false or can't tell: logic with AND and OR	B4 review	9	Input-output rules (function machines)	B5
2	Variables that keep score	B5	10	Design machines: turning symmetry	B4
3	Turtle turns: angles in degrees	B4	11	Tiling and decomposition	B6
4	Inputs and variables: fuel left	B5	12	Timers: variables that change over time	B5
5	Compare and decide	B5	13	Jump rules: loops with conditions	B6
6	Counters and running totals	B5	14	Capstone 1: design a treasure-map game	B7
7	Pattern generators: rules for any step	B4	15	Capstone 2: scores, data and improving the game	B7
8	Search strategies: halve it to find it fast	B6			

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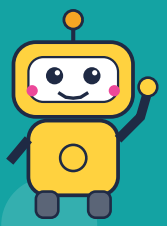
1

CHAPTER 1 · LEVEL B4 review

We the Travellers - I

Coder skill: True, false or can't tell: logic with AND and OR

Goes with NCERT maths, Class 5, chapter 1

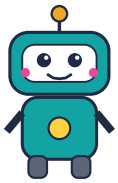


Learn

A computer checks **conditions** that are either **true** or **false**. With what we know, a statement can be true, false, or something we **can't tell** yet.

Two conditions can be joined. With **AND**, both must be true. With **OR**, at least one must be true.

condition	AND	OR
something true or false	both must be true	at least one is true



Before I act, I check: is it true, is it false, or can I not tell yet? And AND means both must be true!

Example

A 3-digit number has only odd digits

1. 'The number is odd.' Its last digit is odd, so this is **definitely true**.
2. 'The number is more than 500.' It could be 135 or 975, so we **can't tell**.
3. 'The number has a 0 in it.' 0 is even, so this is **false**.

★ *Coder fact!* Search engines use AND and OR to find pages that match all, or some, of your words.

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Try it!

Circle the right answer, or write it in the box.

1 A 4-digit number has only even digits. Which statement is **definitely true**?

a) It is even
b) It ends in 0
c) It is more than 5000
d) Its digits add to an odd number

2 Rule: *you may board the bus if you have a ticket AND a mask.* Riya has a ticket but no mask. May she board?

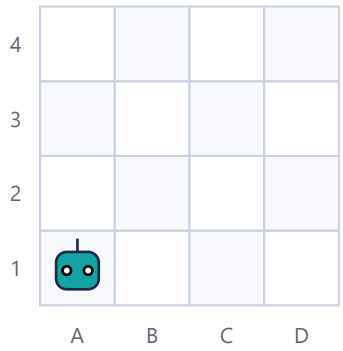
a) yes
b) no

3 Rule: *you get a prize if you score more than 8 OR you finish first.* Arun scored 6 and finished first. Does he get a prize?

a) yes
b) no

4 A code has 3 digits. The middle digit is 2. The first is 1 more than the middle. The last is double the first. What is the code?

5 Robo is on A1 facing North. Program: *repeat 2 times: forward 2, turn right.* Where does Robo end?



a) C3
b) A3
c) C1
d) B2

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Now you build it

Write a checking program using AND.

Ask for a number | If it is even AND more than 50 | say 'Bingo!' | else | say 'Try again'

Test it with 48, 52 and 51. Which numbers win?

Challenge

What is the smallest 5-digit number whose digits are all even and all different?

How did I do?

I can decide true, false or can't tell.	I can use AND and OR.	I can follow a route program.
		

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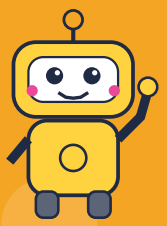
2

CHAPTER 2 · LEVEL B5

Fractions

Coder skill: Variables that keep score

Goes with NCERT maths, Class 5, chapter 2

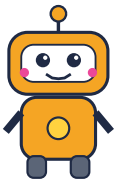


Learn

A **variable** is a named box that stores a number. A program can **change** the number again and again.

Fractions help us change variables: *give away half* means the variable becomes half of what it was.

variable	set	change
a named box for a number	put a value in the box	add or take away from the box



A variable is a named box for a number. My box called 'slices' starts at 8. When I eat a quarter, it changes to 6!

Example

Trace the variable 'slices'

1. Set slices to **8**.
2. Eat one quarter of 8 (that is 2): slices becomes **6**.
3. Eat half of what is left (3): slices becomes **3**.

slices (start)	after 1/4 eaten	after 1/2 of rest
8	6	3

★ *Coder fact!* The battery icon on a phone shows a fraction: how much charge is left.

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Try it!

Circle the right answer, or write it in the box.

1 Find one half of 24 and one third of 24.

2 Which is bigger?

a) 3/4 **b)** 2/3 **c)** same

3 Set water to 20 litres. Use half. Then use half of what is left. What is water now?

4 Which fraction is the same as 2/4?

a) 1/4 **b)** 1/2 **c)** 2/3 **d)** 3/4

5 Sam has 12 marbles. He gives half to Ali, then half of what is left to Bea. How many does Sam keep?

Now you build it

Write a program with a variable called **sweets**.

Set sweets to 20 | Repeat 2 times | give away half of sweets | say sweets

Trace it in a table. What does the program say at the end?

Challenge

A water tank is 3/4 full and holds 60 litres now. How much does it hold when full?

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✓ How did I do?

I can find a fraction of a number.	I can trace a variable.	I can compare fractions.
		

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CHAPTER 3 · LEVEL B4

3 Angles as Turns

Coder skill: Turtle turns: angles in degrees

Goes with NCERT maths, Class 5, chapter 3



Learn

A turn can be measured in **degrees**: a full turn is **360°**, a half turn is **180°** and a quarter turn (a right angle) is **90°**.

Turns can be **clockwise** (right) or **anticlockwise** (left). A drawing robot (a 'turtle') draws shapes with moves and turns: to close any shape it turns 360° in all.

degree a unit for turning	**right angle** a 90 degree turn	**clockwise** the way clock hands turn



A full turn is 360 degrees. Repeat 4 times: forward, turn right 90 degrees. Four quarter turns make a square!

Example

Drawing a square and a triangle

1. Square: *repeat 4 times: forward 50, turn right 90°*. 4 x 90° = 360°.
2. Triangle (equal sides): *repeat 3 times: forward 50, turn right 120°*. 3 x 120° = 360°.
3. Each closed shape needs turns that add up to **360°**.



90° clockwise

★ *Coder fact!* Robot vacuum cleaners turn by set angles to cover a room.

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Try it!

Circle the right answer, or write it in the box.

1 How many degrees are there in a half turn?

2 You face North and turn 90° clockwise. Which way do you face?



a) East **b)** West **c)** South **d)** North

3 You face East and turn 180°. Which way do you face?

a) North **b)** South **c)** West **d)** East

4 What does *repeat 3 times: forward 50, turn right 120°* draw?

a) square **b)** triangle **c)** line **d)** hexagon

5 How big is this turn?



a) 30° **b)** 45° **c)** 90° **d)** 180°

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Now you build it

Write a turtle program that draws a hexagon with 6 equal sides.

Repeat ___ times | forward 30 | turn right ___°

Check: do your turns add up to 360°?

Challenge

To draw a pentagon (5 equal sides), how many degrees must the turtle turn each time?

How did I do?

I can measure turns in degrees.	I can turn clockwise and anticlockwise.	I can draw shapes with turns that add to 360°.
		

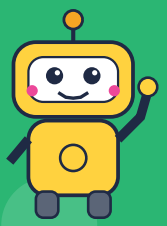
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CHAPTER 4 · LEVEL B5

4 We the Travellers - II

Coder skill: Inputs and variables: fuel left

Goes with NCERT maths, Class 5, chapter 4

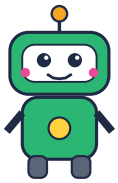


Learn

An **input** is information given to a program while it runs, like 'How far will you travel?'.

The program uses the input to **update a variable**. A car's computer does this to show how much fuel is left.

input	output	update
information given to a program	what the program gives back	change a variable's value



An input is information the program asks for, like the distance of a trip. I use it to update my 'fuel' variable!

Example

How much fuel is left?

1. Set fuel to **40** litres. The car uses 8 litres for every 100 km.
2. Input: the trip is **300 km**, so the car uses 3 x 8 = 24 litres.
3. Update: fuel = 40 - 24 = **16 litres** left.

fuel	input: km	fuel left
40	300	16

★ *Coder fact!* Cars work out 'distance to empty' by doing a fuel calculation again and again.

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Try it!

Circle the right answer, or write it in the box.

1 Fuel is 40 litres and the car uses 8 litres per 100 km. After a 200 km trip, how much fuel is left?

2 At a fair, 3 stickers swap for 1 badge. Meena has 10 stickers and swaps as many as she can. How many badges and stickers does she have?

3 A program asks for your age. *If age is less than 13, say 'Child', else say 'Teen or adult'*. What does it say for 13?

a) Child **b)** Teen or adult

4 Delhi to Agra is 230 km and Agra to Jaipur is 240 km. How far is Delhi to Jaipur through Agra?

5 3 buses have 45 seats each. 120 travellers get on. How many seats are empty?

Now you build it

Write a fuel program that uses an input.

Set fuel to 40 **Ask: how many km?** **set used to km / 100 x 8**

change fuel by - used **If fuel is less than 5** **say 'Fill up soon!'**

Run it with 100 km, then 450 km. What does it say each time?

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★ Challenge

The car uses 8 litres per 100 km and the tank holds 40 litres. What is the longest trip on one full tank?

✓ How did I do?

I can use an input in a program.	I can update a variable.	I can solve travel problems step by step.
		

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CHAPTER 5 · LEVEL B5

5 Far and Near

Coder skill: Compare and decide

Goes with NCERT maths, Class 5, chapter 5

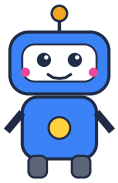


Learn

Programs **compare** numbers with **>** (more than), **<** (less than) and **=** (equal), then decide what to do.

To find the nearest place, keep a variable for the **smallest distance so far**. Check each place: if it is closer, update the variable.

compare check which is bigger	**less than (<)** smaller than	**kilometre** 1,000 metres



To find the nearest shop, I keep a variable called 'nearest'. If a new shop is closer, I change it!

Example

Walk, cycle or bus?

1. Rule: *if distance < 5 km then cycle, else take the bus.*
2. Asha lives 2 km away: cycle. Bilal lives 7 km away: bus. Charu lives 4 km away: cycle.
3. The same rule makes a different choice for each input.

★ *Coder fact!* Food delivery apps compare distances to send the nearest delivery partner.

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Try it!

Circle the right answer, or write it in the box.

1 How many metres are there in 2 km 300 m?

2 Which is the nearest?

a) 1.5 km **b)** 900 m **c)** 2 km

3 Rule: *if distance > 3 km then take the bus, else walk*. The distance is exactly 3 km. What do you do?

a) bus **b)** walk

4 A charger cable is 1 m long. The socket is 2 m from the bed. Can you charge your phone while sitting on the bed?

a) yes **b)** no

5 Order from nearest to farthest: 1,200 m; 1 km 50 m; 980 m.

Now you build it

Write a 'nearest shop' program that uses a variable called **nearest**.

Set nearest to the first shop's distance | For each other shop | If its distance < nearest | set nearest to its distance

Trace it with shops at 400 m, 250 m and 600 m. What is nearest at the end?

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★ Challenge

Ravi walks 3 km in 1 hour. The bus covers 5 km in 15 minutes. School is 5 km away. How much faster is the bus?

✓ How did I do?

I can compare distances.	I can choose with an if-else rule.	I can find the nearest with a variable.
		

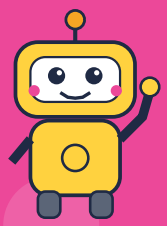
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CHAPTER 6 · LEVEL B5

6 The Dairy Farm

Coder skill: Counters and running totals

Goes with NCERT maths, Class 5, chapter 6

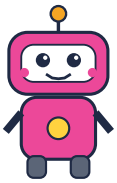


Learn

A **counter** is a variable that goes up by 1 each time something happens, like counting cows.

A **running total** is a variable that adds each new amount: *total* = *total* + *milk*. Start it at 0 before the loop.

counter	total	litre
goes up by 1 each time	adds up amounts	1,000 millilitres



My 'total' box starts at 0. For each cow, I add her milk. At the end, the box holds the whole day's milk!

Example

Milk from 5 cows

1. Set total to 0. The cows give 8, 10, 7, 9 and 6 litres.
2. Add each one: 8, then 18, then 25, then 34, then **40**.
3. The farm collected **40 litres**. A counter would show 5 cows.

Cow	Milk (litres)	total
start		0
1	8	8
2	10	18
3	7	25
4	9	34
5	6	40

★ *Coder fact!* Many dairy cooperatives in India record each farmer's milk on machines that keep running totals.

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Try it!

Circle the right answer, or write it in the box.

1 Find the running total of 12, 15, 9 and 14.

2 Each full circle means 10 litres of milk. How many litres did the buffaloes give?



3 Count the days with more than 30 litres: 28, 35, 31, 30, 40.

4 How many millilitres are in 2.5 litres?

5 40 litres of milk are packed into 500 ml packets. How many packets?

Now you build it

Write a milk-counting program with a counter and a total.

Set total to 0 | Set cows to 0 | For each cow | change total by her milk | change cows by 1 | say total and cows

Why must total start at 0 and not at 10?

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★ Challenge

The farm sells milk at Rs 50 a litre and collects 40 litres a day. How much does it earn in a week?

✓ How did I do?

I can use a counter.	I can keep a running total.	I can read a picture graph with half symbols.
		

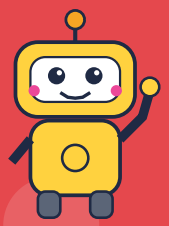
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CHAPTER 7 · LEVEL B4

7 Shapes and Patterns

Coder skill: Pattern generators: rules for any step

Goes with NCERT maths, Class 5, chapter 7

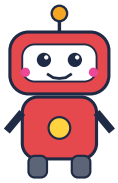


Learn

A **growing pattern** follows a rule. If you find the rule, you can work out **any step** without drawing all the steps before it.

A rule like *sticks = 3 x squares + 1* is a **pattern generator**. Spreadsheets and programs use rules like this all the time.

rule	step	generator
tells how a pattern grows	one stage of a pattern	a rule that makes the pattern



Why build 100 steps of a pattern? If I know the rule, I can jump straight to step 100!

Example

Matchstick squares in a row

1. 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10.
2. Each new square adds 3 sticks. Rule: **sticks = 3 x squares + 1**.
3. So 10 squares need 3 x 10 + 1 = **31 sticks**.

★ *Coder fact!* Spreadsheets use rules called formulas to fill a whole column from one pattern.

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Try it!

Circle the right answer, or write it in the box.

1 What comes next: **4, 7, 10, 13, ?**

2 Use the matchstick rule. How many sticks do 20 squares need?

3 This pattern grows: 1, 4, 9 dots. How many dots in step 4?



step 1 step 2 step 3 step 4

4 Which rule makes 3, 6, 9, 12 for steps 1, 2, 3, 4?

a) step + 3 **b)** step x 3 **c)** step x 2 + 1 **d)** step + 2

5 A shape pattern repeats: triangle, circle, square, star. What is the 17th shape?

a) triangle **b)** circle **c)** square **d)** star

Now you build it

Turn this rule into a table: **tiles = step x 2 + 1**.

Repeat for steps 1 to 5 | set tiles to step x 2 + 1 | write step and tiles

Fill the table, then draw step 3.

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★ Challenge

In the matchstick pattern, which step uses 61 sticks? Which uses 100?

✓ How did I do?

I can continue a growing pattern.	I can find a pattern's rule.	I can use a rule to jump to any step.
		

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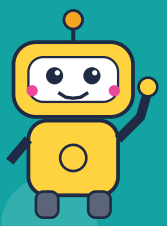
CHAPTER 8 · LEVEL B6

8

Weight and Capacity

Coder skill: Search strategies: halve it to find it fast

Goes with NCERT maths, Class 5, chapter 8

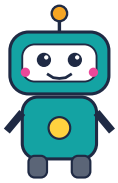


Learn

A **one-by-one search** (linear search) checks things in order until it finds the right one. It can take a long time.

A **halving search** (binary search) checks the middle and throws away the half that cannot hold the answer. For 1 to 32 it needs at most 5 yes/no questions. Computers use it to find things very fast.

linear search	halving search	strategy
check one by one	check the middle, drop half	a plan for solving



To guess a number from 1 to 32, I always pick the middle. Each guess throws away half the numbers!

Example

Guess my number (1 to 32)

1. Guess 16: 'higher'. Now only 17 to 32 are left.
2. Guess 24: 'lower'. Now only 17 to 23 are left (grey numbers are ruled out).
3. Keep guessing the middle of what is left. You will find it in a few more guesses.



★ *Coder fact!* Phone and dictionary apps find one word among millions in a few steps by halving the search.

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Try it!

Circle the right answer, or write it in the box.

1 One of 8 coins is heavier. You use a balance and split the coins in half each time. How many weighings do you need?



a) 2 **b)** 3 **c)** 4 **d)** 8

2 You look for a name on 20 cards one by one. What is the most cards you might check?

3 How many 'Is it more than ...?' questions do you need, at most, to find a number from 1 to 16 by halving?

4 Look at the example strip. Which numbers are still possible?

5 A 2 kg bag is filled with 250 g packets. How many packets?

Now you build it

Write the 'guess my number' program using halving.

Set low to 1 and high to 32 **Repeat until found** **guess the middle of low and high**

If too small: set low to guess + 1 **If too big: set high to guess - 1**

Play with a friend. Count your guesses. Is it ever more than 6?

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★ Challenge

One of 9 coins is heavier. What is the fewest weighings on a balance to be sure which one?

✓ How did I do?

I can search one by one.	I can search by halving.	I can explain which strategy is faster.
		

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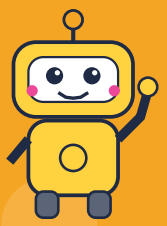
9

CHAPTER 9 · LEVEL B5

Coconut Farm

Coder skill: Input-output rules (function machines)

Goes with NCERT maths, Class 5, chapter 9

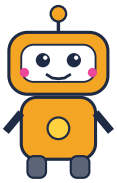


Learn

A **function machine** takes an **input**, follows a **rule**, and gives an **output**.

If you know some inputs and outputs, you can work out the hidden rule. Machines can be **chained**: the output of one becomes the input of the next.

function machine	rule	chain
input, rule, output	what the machine does	join machines one after another



Put a number into my machine, I follow my rule, and out comes the answer. Can you guess my secret rule?

Example

The harvest calculator

1. Each coconut tree gives 12 coconuts. Rule: **x 12**.
2. Input: 6 trees. Output: 6 x 12 = **72 coconuts**.
3. Change the input and the same rule gives a new output.



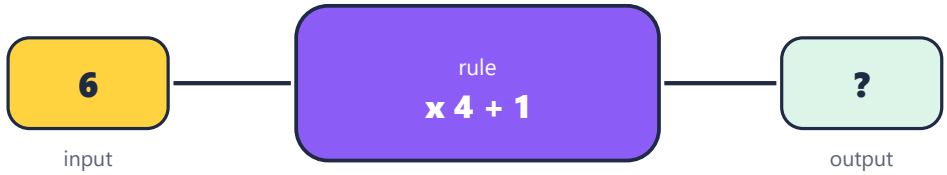
★ *Coder fact!* Every key on a calculator is a small rule machine: input in, rule applied, output out.

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Try it!

Circle the right answer, or write it in the box.

1 What comes out of this machine?



2 Find the rule: 2 gives 10, 3 gives 15, 5 gives 25.

a) + 8 **b)** x 5 **c)** x 2 + 6 **d)** + 20

3 Using the rule **x 4 + 1**, which input gives an output of 49?

4 A farm has 25 trees with 30 coconuts each. How many coconuts in all?

5 The 750 coconuts are packed 12 to a bag. How many full bags, and how many coconuts are left over?

Now you build it

Design your own two-step machine and test it on a friend.

Ask for an input **step 1: ___** **step 2: ___** **say the output**

Give your friend three inputs and outputs. Can they find your rule?

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★ Challenge

Chain two machines: **x 2** then **+ 3**. What does 7 give? Is it the same if you do **+ 3** first, then **x 2**?

✓ How did I do?

I can use an input-output rule.	I can find a hidden rule.	I can chain two machines.
		

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CHAPTER 10 · LEVEL B4

10

Symmetrical Designs

Coder skill: Design machines: turning symmetry

Goes with NCERT maths, Class 5, chapter 10



Learn

A design has **rotational symmetry** if it looks the same after turning it less than a full turn. The number of matching positions is its **order**.

Design programs use loops: *repeat n times: draw a petal, turn 360 / n degrees*. The turns must add up to 360°.

rotational symmetry	order	reflection
looks the same after a turn	how many matching positions	a mirror image

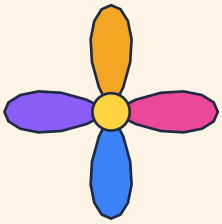


Repeat 6 times: draw a petal, turn 60 degrees. 6 x 60 = 360, so my flower closes perfectly!

Example

A 4-petal flower

1. Program: *repeat 4 times: draw a petal, turn 90°*.
2. 4 x 90° = 360°, so the last petal meets the first.
3. Turn it by a quarter turn and it looks the same: order **4**.



★ *Coder fact!* Kolam and rangoli artists use repeat-and-turn rules, much like coders do.

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Try it!

Circle the right answer, or write it in the box.

1 What is the order of rotational symmetry of this 6-petal flower?



2 To draw 6 equal petals, how much should the program turn after each petal?

a) 30° **b)** 45° **c)** 60° **d)** 90°

3 Which letters look the same after a half turn?

a) N only **b)** S only **c)** Z only **d)** N, S and Z

4 How many lines of symmetry does a triangle with 3 equal sides have?

5 A design has 8 equal petals. What angle is each turn?

Now you build it

Write a rangoli program for a design with 5 petals.

Repeat ___ times | draw a petal | turn ___°

Draw the design and colour it so it still has rotational symmetry.

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★ Challenge

A program turns 40° after each petal. How many petals until the design closes?

✓ How did I do?

I can find rotational symmetry.	I can work out the turn for n petals.	I can write a design program.
		

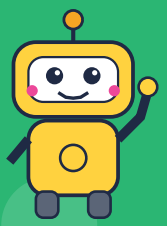
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CHAPTER 11 · LEVEL B6

11 Grandmother's Quilt

Coder skill: Tiling and decomposition

Goes with NCERT maths, Class 5, chapter 11

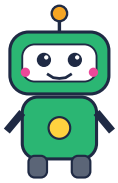


Learn

Decomposition means breaking a big problem into small parts. A big quilt can be broken into one small **tile** that repeats.

To find an **area**, count square units. A square cut along its diagonal makes two triangles, each half a square unit.

tile	area	decompose
a part that repeats	how many square units cover a shape	break into parts

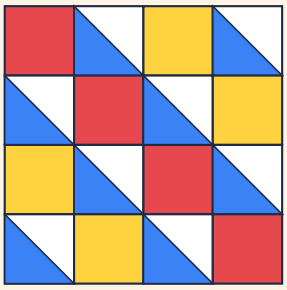


A big quilt is just one small tile repeated. Find the tile, and the whole design is easy to plan!

Example

Grandmother's quilt

1. The quilt is 4 squares by 4 squares: 16 squares in all.
2. Split it into four 2 x 2 blocks. Two blocks are red-and-blue and two are yellow-and-blue.
3. They are laid out like a chessboard. So the whole quilt comes from **one 2 x 2 tile in two colours**.



★ *Coder fact!* Video games build big worlds out of small tiles that repeat, just like a quilt.

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Try it!

Circle the right answer, or write it in the box.

1. In the example quilt, how many solid red squares are there?

2. How many squares have a blue half-triangle?

3. What is the total blue area, in whole squares?

4. A quilt is 6 squares long and 4 squares wide. How many squares in all?

5. How many 2 x 2 tiles cover a 6 x 4 quilt?

Now you build it

Design a 2 x 2 tile, then write a nested loop to repeat it across a 4 x 4 quilt.

Repeat 2 times (rows of tiles) **Repeat 2 times (tiles in a row)** **place my tile** **move to the next space**

Draw your quilt on squared paper. How many tiles did you place?

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★ Challenge

An 8 x 8 quilt is made of 2 x 2 tiles, and each tile has exactly 1 red square. How many red squares are in the quilt?

✓ How did I do?

I can find the tile that repeats.	I can count area, including halves.	I can plan a design by decomposition.
		

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CHAPTER 12 · LEVEL B5

12

Racing Seconds

Coder skill: Timers: variables that change over time

Goes with NCERT maths, Class 5, chapter 12

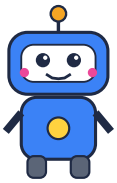


Learn

A **timer** is a variable that changes every second: *timer = timer + 1*. A **countdown** goes the other way: *timer = timer - 1*.

1 minute = 60 seconds. In a race, the **smallest** time wins.

timer	countdown	second
counts seconds	counts back to zero	60 make a minute



My 'timer' variable goes up by 1 every second. When it reaches 60, one minute has passed!

Example

Who won the race?

1. Ana finished in 14 seconds, Bo in 12 seconds and Cy in 15 seconds.
2. Compare the times: the smallest is 12.
3. **Bo** won. Ana was 2 seconds behind.



★ *Coder fact!* Race timers use electronic sensors to measure times to a hundredth of a second.

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Try it!

Circle the right answer, or write it in the box.

1 How many seconds are there in 2 minutes 15 seconds?

2 Which time is the fastest?

a) 48 s **b)** 1 min 5 s **c)** 52 s

3 A timer starts at 0 and adds 1 every second. A song lasts 3 minutes. What does the timer show at the end?

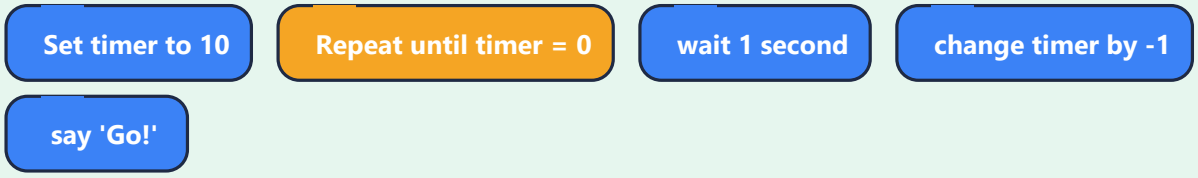
4 4 runners in a relay each run for 15 seconds. How long is the whole relay?

a) 45 s **b)** 1 minute **c)** 1 min 15 s **d)** 40 s

5 A countdown starts at 10 and takes away 1 every second. How many seconds until it shows 0?

Now you build it

Write a countdown program for a race start.



Act it out: one friend is the timer and says each number. Does 'Go!' come after 0?

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★ Challenge

Ria runs 100 m in 20 seconds. If she keeps the same speed, how long will 400 m take?

✓ How did I do?

I can change minutes into seconds.	I can compare race times.	I can write a countdown.
		

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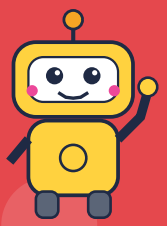
13

CHAPTER 13 · LEVEL B6

Animal Jumps

Coder skill: Jump rules: loops with conditions

Goes with NCERT maths, Class 5, chapter 13

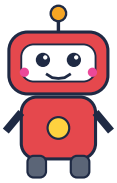


Learn

An animal that always jumps the same distance lands on **multiples** of that number: a frog jumping 3 lands on 3, 6, 9, 12, ...

Two animals both land on **common multiples**. A loop with a condition can find them: *repeat until position > 30: jump; if position is a multiple of 4, say 'Meet!'*

multiple	common multiple	repeat until
a number in a times table	in both times tables	loop until a condition is true

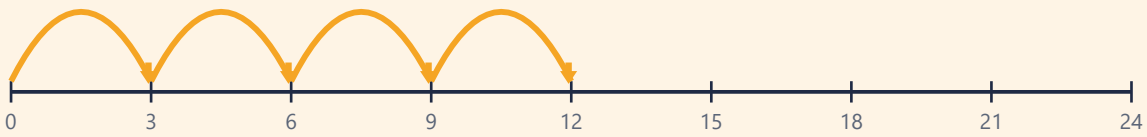


Frog jumps 3, rabbit jumps 4. Where do we both land? On the common multiples: 12, 24, 36!

Example

Frog (3) and rabbit (4)

1. Frog lands on 3, 6, 9, **12**, 15, 18, 21, **24**, ...
2. Rabbit lands on 4, 8, **12**, 16, 20, **24**, ...
3. They first land on the same spot at **12**.



★ *Coder fact!* A bus every 15 minutes and a bus every 20 minutes leave together every 60 minutes: a common multiple!

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Try it!

Circle the right answer, or write it in the box.

1. A kangaroo jumps 5 each time from 0. Will it land on 35?

   a) yes **b)** no

2. Frog jumps 3, grasshopper jumps 4, both from 0. Where do both land, up to 30?

3. *Start at 0. Repeat until position > 20: jump 6.* Where does it stop?

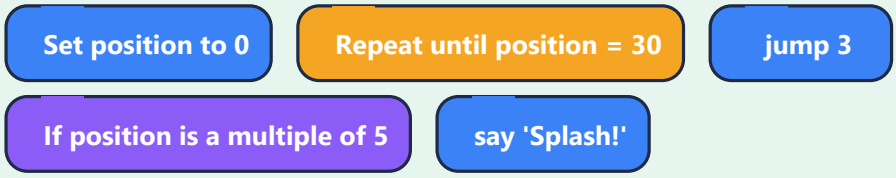
4. Does a frog jumping 4 each time land on 30?

   a) yes **b)** no

5. A rabbit jumps 7 each time from 0. Where is it after 10 jumps?

Now you build it

Write a jump game program with a condition inside the loop.



On which positions does the program say 'Splash!'?

Challenge

Two frogs start at 0. One jumps 6 and the other jumps 8. Where do they first land together?

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✓ How did I do?

I can skip count with jumps.	I can find common landings.	I can use a loop with a condition.
		

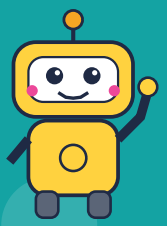
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CHAPTER 14 · LEVEL B7

14 Maps and Locations

Coder skill: Capstone 1: design a treasure-map game

Goes with NCERT maths, Class 5, chapter 14

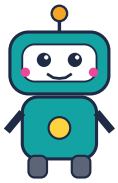


Learn

Maps use **grid references** (like C3) and **compass directions** (North, South, East, West). A **scale** tells real distance: 1 square = 100 m.

A good game has a **goal**, a **start**, **rules** and a way to **win**. Designing one uses everything you have learnt: routes, conditions, variables and testing.

grid reference	scale	game design
a letter and number for a square	map distance to real distance	planning goal, rules and winning

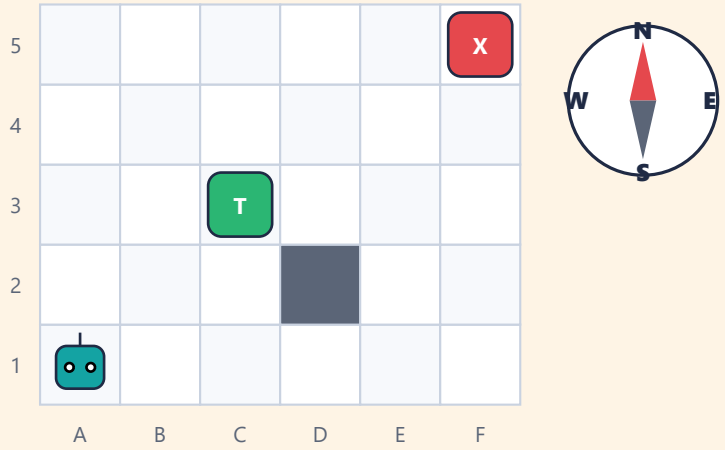


Let's make our own game! A map, a start, a treasure, some walls and clear rules. Then players write programs to win.

Example

The treasure map

1. Start at A1. The treasure X is at F5. The grey square is a wall. The tree T is at C3.
2. One route: North 4, then East 5. That is 9 moves.
3. Scale: 9 squares x 100 m = 900 m of walking.



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Try it!

Circle the right answer, or write it in the box.

1 On the example map, what is at C3?

a) the treasure **b)** a tree **c)** a wall **d)** nothing

2 From A1, go 4 squares East and 4 squares North. Which square are you on?

3 1 square = 100 m. How far is it from A1 to F1 along the bottom row?

4 Which direction is the opposite of North-East?

a) South-East **b)** South-West **c)** North-West **d)** West

5 What is the fewest number of moves from A1 to the treasure at F5 on the example map?

Now you build it

Design your own treasure-map game on a 5 x 6 grid.

Choose a start square **Hide the treasure** **Add 2 walls and 1 bonus square**

Write the rules **Test it yourself**

Draw your map and write your rules. Test it: can the treasure always be reached?

Challenge

Write the shortest program for your own map. Ask a friend to find an even shorter one.

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✓ How did I do?

I can use grid references and directions.	I can use a map scale.	I can design a game with clear rules.
		

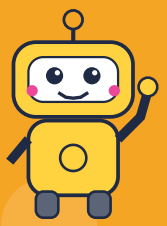
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CHAPTER 15 · LEVEL B7

15 Data Through Pictures

Coder skill: Capstone 2: scores, data and improving the game

Goes with NCERT maths, Class 5, chapter 15

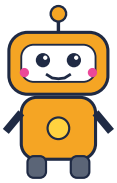


Learn

A game keeps **score** in a variable: *add 10 for the treasure, take 2 for bumping a wall.*

When friends play, collect the results as **data** and show them in a **picture graph** or **bar graph**. The data shows what to improve: was the game too easy, too hard or unfair?

score	pictograph	test
a variable for points	a graph with pictures	try it to find problems

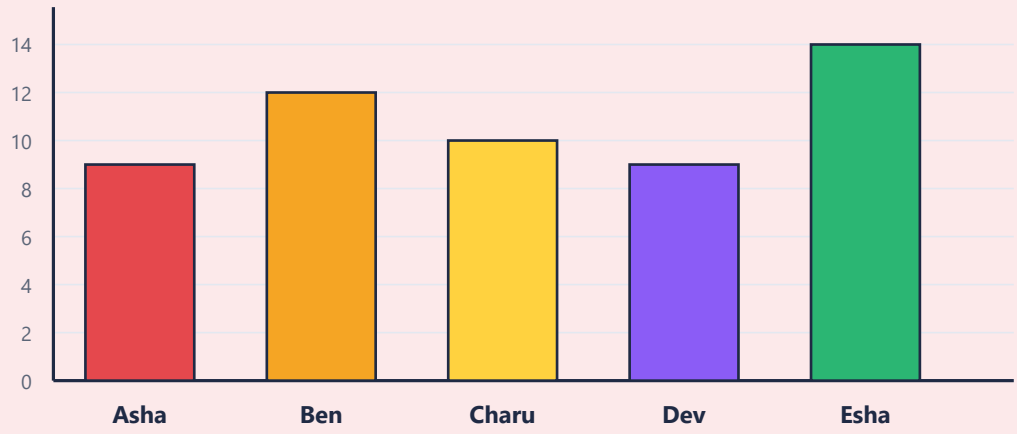


Now we finish the game: add a score, play it with friends, and use the data to make it better!

Example

Scores from the treasure game

1. Moves to the treasure: Asha 9, Ben 12, Charu 10, Dev 9, Esha 14.
2. Fewest moves wins, so Asha and Dev **tie** with 9.
3. A tie-break rule is needed: fewest wall bumps wins.



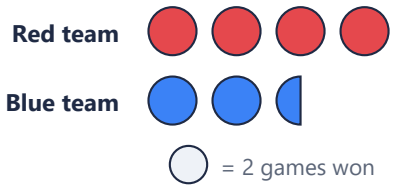
★ *Coder fact!* Game makers study players' data to make levels more fun and fair.

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Try it!

Circle the right answer, or write it in the box.

1 Each symbol means 2 games won. The red team has 4 symbols. How many games did they win?



2 From the same picture graph, how many games did the blue team win?

3 Score rule: add 10 for each treasure, take 2 for each wall bump. A player finds 3 treasures and bumps 2 walls. What is the score?

4 Tie-break: *if two players have the same moves, then fewer wall bumps wins.* Asha: 9 moves, 1 bump. Dev: 9 moves, 0 bumps. Who wins?

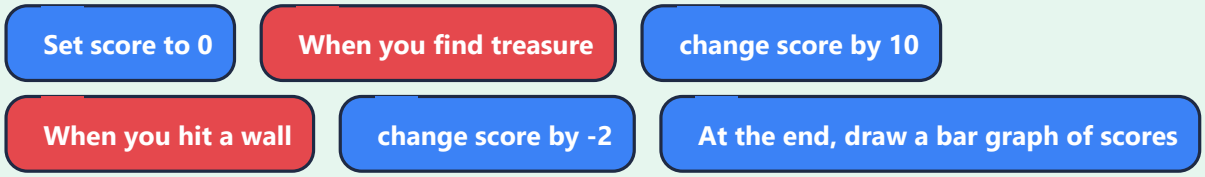
a) Asha **b)** Dev **c)** tie

5 In the bar graph, how many more moves did Esha take than Asha?

Now you build it

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Finish your game from chapter 14 by adding a score.



Play with 4 friends, draw a graph of the scores and write one change that would make the game better.

★ Challenge

Design a fairer scoring rule for your game and explain why it is fairer.

✓ How did I do?

I can read picture graphs and bar graphs.	I can keep score with a variable.	I can use data to improve a game.
		

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Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. We the Travellers - I

1. a) It is even
2. b) no: both must be true
3. a) yes: one is enough with OR
4. 326
5. a) C3 (facing South)
Build it: Only 52 (48 is not more than 50; 51 is not even)
Challenge: 20,468 (0 cannot come first)

2. Fractions

1. 12 and 8
2. a) 3/4
3. 5 litres
4. b) 1/2
5. 3
Build it: 20, then 10, then 5: it says 5
Challenge: 80 litres (1/4 is 20 litres)

3. Angles as Turns

1. 180
2. a) East
3. c) West
4. b) triangle
5. b) 45°
Build it: Repeat 6 times: forward 30, turn right 60° (6 x 60 = 360)
Challenge: 72° (360 / 5)

4. We the Travellers - II

1. 24 litres
2. 3 badges and 1 sticker
3. b) Teen or adult
4. 470 km
5. 15
Build it: 100 km: fuel 32, no warning; 450 km: fuel 4, says 'Fill up soon!'
Challenge: 500 km

5. Far and Near

1. 2,300 m
2. b) 900 m
3. b) walk (3 is not more than 3)
4. b) no
5. 980 m; 1 km 50 m; 1,200 m
Build it: 400, then 250, then stays 250: nearest is 250 m
Challenge: 85 minutes faster (walking takes 1 hour 40 minutes; the bus takes 15 minutes)

6. The Dairy Farm

1. 50
2. 25 litres
3. 3
4. 2,500 ml
5. 80
Build it: If it starts at 10, every answer is 10 too big
Challenge: Rs 14,000

7. Shapes and Patterns

1. 16
2. 61
3. 16 (4 x 4)
4. b) step x 3
5. a) triangle (17 = 4 x 4 + 1)
Build it: Steps 1-5: 3, 5, 7, 9, 11 tiles
Challenge: Step 20 uses 61 sticks; step 33 uses 100 (3 x 33 + 1 = 100)

8. Weight and Capacity

1. b) 3 (4 vs 4, then 2 vs 2, then 1 vs 1)
2. 20
3. 4
4. 17 to 23
5. 8
Build it: No: halving finds any number from 1 to 32 in at most 6 guesses (5 yes/no splits)
Challenge: 2: weigh 3 against 3; the heavy group (or the 3 left out) is known; then weigh 1 against 1 from that group

9. Coconut Farm

1. 25
2. b) x 5
3. 12
4. 750
5. 62 bags, 6 left over
Build it: Open answer
Challenge: 17; no: + 3 first gives 20, so the order matters

10. Symmetrical Designs

1. 6
2. c) 60°
3. d) N, S and Z
4. 3
5. 45°
Build it: Repeat 5 times: draw a petal, turn 72°
Challenge: 9 (9 x 40 = 360)

11. Grandmother's Quilt

1. 4
2. 8
3. 4 square units (8 halves)
4. 24
5. 6
Build it: 4 tiles
Challenge: 16 (there are 16 tiles)

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12. Racing Seconds

1. 135
2. a) 48 s
3. 180
4. b) 1 minute
5. 10
Build it: Yes: the loop stops at 0, then the program says 'Go!'
Challenge: 80 seconds (1 minute 20 seconds)

13. Animal Jumps

1. a) yes (7 jumps)
2. 12 and 24
3. 24
4. b) no (30 is not a multiple of 4)
5. 70
Build it: 15 and 30
Challenge: 24

14. Maps and Locations

1. b) a tree
2. E5
3. 500 m
4. b) South-West
5. 9
Build it: Open answer
Challenge: Open answer

15. Data Through Pictures

1. 8
2. 5
3. 26
4. b) Dev
5. 5
Build it: Open answer
Challenge: Open answer, for example: give bonus points for the shortest route so players plan instead of rushing

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Coder words

AND	both must be true	Ch 1
area	how many square units cover a shape	Ch 11
chain	join machines one after another	Ch 9
change	add or take away from the box	Ch 2
clockwise	the way clock hands turn	Ch 3
common multiple	in both times tables	Ch 13
compare	check which is bigger	Ch 5
condition	something true or false	Ch 1
countdown	counts back to zero	Ch 12
counter	goes up by 1 each time	Ch 6
decompose	break into parts	Ch 11
degree	a unit for turning	Ch 3
function machine	input, rule, output	Ch 9
game design	planning goal, rules and winning	Ch 14
generator	a rule that makes the pattern	Ch 7
grid reference	a letter and number for a square	Ch 14
halving search	check the middle, drop half	Ch 8
input	information given to a program	Ch 4
kilometre	1,000 metres	Ch 5
less than (<)	smaller than	Ch 5
linear search	check one by one	Ch 8
litre	1,000 millilitres	Ch 6
multiple	a number in a times table	Ch 13
OR	at least one is true	Ch 1
order	how many matching positions	Ch 10
output	what the program gives back	Ch 4
pictograph	a graph with pictures	Ch 15
reflection	a mirror image	Ch 10
repeat until	loop until a condition is true	Ch 13
right angle	a 90 degree turn	Ch 3
rotational symmetry	looks the same after a turn	Ch 10
rule	tells how a pattern grows	Ch 7
scale	map distance to real distance	Ch 14
score	a variable for points	Ch 15
second	60 make a minute	Ch 12
set	put a value in the box	Ch 2
step	one stage of a pattern	Ch 7

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strategy	a plan for solving	Ch 8
test	try it to find problems	Ch 15
tile	a part that repeats	Ch 11
timer	counts seconds	Ch 12
total	adds up amounts	Ch 6
update	change a variable's value	Ch 4
variable	a named box for a number	Ch 2

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

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